Combinations (medium)

Problem Statement

Given two integers n and k, return all possible combinations of k numbers chosen from the range `[1, n]`.

Each combination must be a unique set of numbers, order of which does not matter.

Examples

- **Example 1**:
  - **Input**: `n = 3, k = 2`
  - **Expected Output**: `[[1, 2], [1, 3], [2, 3]]`
  - **Justification**: `[[1, 2], [1, 3], [2, 3]]` are all combinations of size 2, which we can create using `[1, 2, 3]`.
- **Example 2**:
  - **Input**: `n = 4, k = 1`
  - **Expected Output**: `[[1], [2], [3], [4]]`
  - **Justification**: `[[1], [2], [3], [4]]` are all combinations of size 1, which we can create using `[1, 2, 3, 4].
- **Example 3**:
  - **Input**: `n = 5, k = 3`
  - **Expected Output**: `[[1, 2, 3], [1, 2, 4], [1, 2, 5], [1, 3, 4], [1, 3, 5], [1, 4, 5], [2, 3, 4], [2, 3, 5], [2, 4, 5], [3, 4, 5]]`
  - **Justification**: We unique triplets using numbers 1 to 5, yielding ten different combinations.

Try it yourself

Try solving this question here:

Java

```java
import java.util.ArrayList;
import java.util.List;

public class Solution {
    // Function to generate all combinations
    public List<List<Integer>> combine(int n, int k) {
        List<List<Integer>> result = new ArrayList<>();
        // ToDo: Write Your Code Here.
        return result;
    }
}
```

Test Case | Run Code Result | Stdout

```
3
2
```

Use Example Testcases | Run | Submit